CÓDIGO:

```
import numpy as np from numpy.random import randn

N, D_in, H, D_out = 64, 1000, 100, 10 x, y = randn(N, D_in),
randn(N, D_out) w1, w2 = randn(D_in, H), randn(H, D_out)

for t in range(2000):

h = 1 / (1 + np.exp(-x.dot(w1)))y_pred = h.dot(w2)loss =
np.square(y_pred - y).sum()if t % 100 == 0:    print(f"iter {t:4d} |
loss = {loss:.6f}")grad_y_pred = 2.0 * (y_pred - y)grad_w2 =
h.T.dot(grad_y_pred)grad_h = grad_y_pred.dot(w2.T)grad_w1 =
x.T.dot(grad_h * h * (1 - h))w1 -= 1e-4 * grad_w1w2 -= 1e-4 *
grad_w2

print(f"loss final = {loss:.6f}")
```

SAÍDA DO CÓDIGO :

